PROBLEM VF9

Conveyor Belt Pumping on an Incline

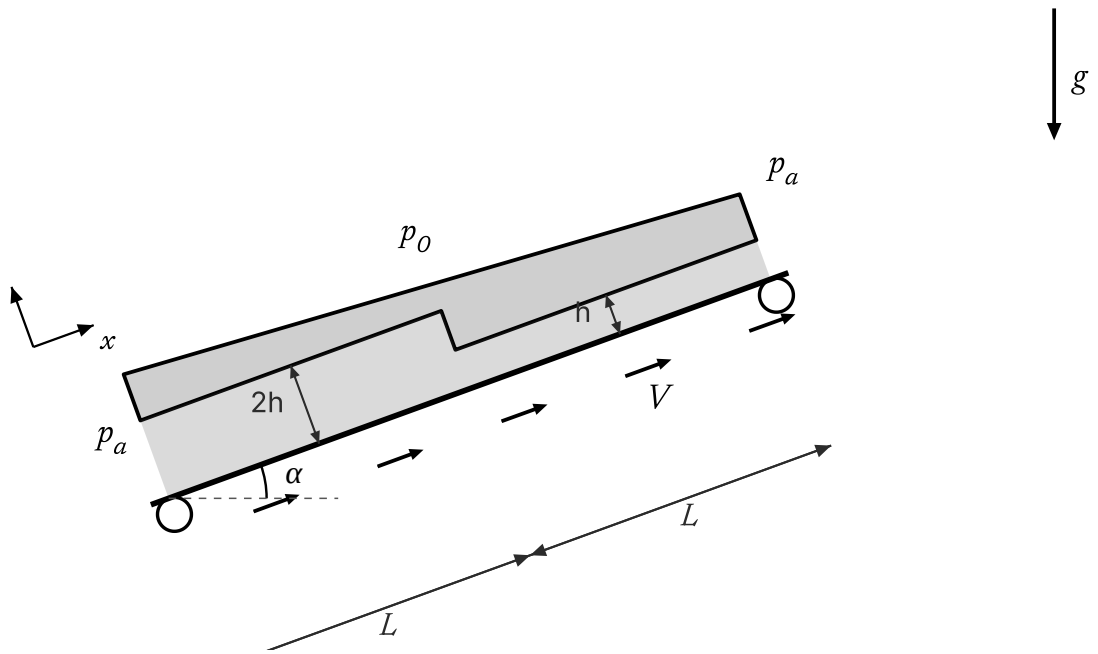
Original statement · VF9.pdf

Couette + Poiseuille | Two stretches in series | Power and energy balance

OFFICIAL PROBLEM STATEMENT

The conveyor belt of velocity V shown in the figure is used to pump a liquid of density ρ and viscosity μ along a straight inclined channel of angle α including two stretches of equal length L with different heights $2h$ and h, respectively.

- Give the conditions for the liquid-film motion to be dominated by viscosity.
- Compute the pressure distribution along the channel $p(x)$.
- Determine the volume flux Q as well as the minimum value of V for which $Q > 0$.
- Obtain the power needed to drive the conveyor belt.



Statement figure (redrawn from VF9.pdf): an inclined channel at the angle α between the conveyor belt, which moves up the slope with velocity V, and a fixed wall with a step (gap 2h over a length L, then h over a length L); both ends are at p_a and p_0 is the pressure at the step

`Source: VF9.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.8-7.12; wedge effect, Eqs. 7.69-7.75.`

Worked solution

Hypotheses, Data and Unknowns

- The belt is the lower wall of the channel (the line between the two rollers in the figure) and moves up the slope, along $+x$, with velocity V; the upper wall is fixed. x runs along the belt and y is normal to it, from the belt towards the fixed wall.
- Gap: $2h$ for $0 < x < L$ and h for $L < x < 2L$. Both ends of the channel are at the atmospheric pressure p_a (the inlet at the bottom reservoir, the outlet at the top).
- The flow is steady and slender, so in each stretch the pressure is uniform across the gap and the reduced pressure $P = p + \rho g x \sin\alpha$ (with $Z = x\sin\alpha$ at $y \to 0$) has a constant gradient.
- Let p_0 be the pressure at the step, $x = L$.

Symbol	Meaning	SI unit
V	velocity of the belt	m/s
h, $2h$	gaps of the two stretches	m
L, α	length of each stretch, angle of the channel	m, rad
p_a, p_0	pressure at the ends and at the step	Pa
Q	volume flux per unit width	m^2/s

Criterion for Viscous Dominance

For a slender film (Eqs. 7.67-7.68) with the velocity scale V, the largest gap $2h \sim h$ and the length L:

$$\frac{\rho V h}{\mu}\,\frac{h}{L} \ll 1$$

The flow is steady, so the unsteady condition (7.67) holds.

Velocity and Flux in Each Stretch

The reduced pressure at the ends: $P(0) = p_a$ and $P(2L) = p_a + \rho g\, 2L \sin\alpha$; at the step $P(L) = p_0 + \rho g L \sin\alpha$. With $P_l = -dP/dx$:

$$P_{l1} = \frac{p_a - p_0 - \rho g L \sin\alpha}{L}, \qquad P_{l2} = \frac{p_0 + \rho g L \sin\alpha - p_a - 2\rho g L \sin\alpha}{L} = \frac{p_0 - p_a - \rho g L \sin\alpha}{L}$$

The profile of a gap H with the lower wall at velocity V and the upper wall at rest is the sum of Couette and Poiseuille flows (Eqs. 7.9 and 7.11, 7.70):

$$v_x = V\,\frac{H - y}{H} + \frac{P_l}{2\mu}\, y\,(H - y), \qquad Q = \int_0^H v_x\, dy = \frac{V H}{2} + \frac{P_l H^3}{12\mu}$$

$$H = 2h: \quad Q = Vh + \frac{2h^3}{3\mu} P_{l1}, \qquad H = h: \quad Q = \frac{Vh}{2} + \frac{h^3}{12\mu} P_{l2}$$

Pressure Distribution and Flux

The flux is the same in both stretches. With $\Delta = (p_0 - p_a)/L$ and $G = \rho g \sin\alpha$, $P_{l1} = -\Delta - G$ and $P_{l2} = \Delta - G$:

$$Vh - \frac{2h^3}{3\mu}(\Delta + G) = \frac{Vh}{2} + \frac{h^3}{12\mu}(\Delta - G)$$

Multiplying by $12\mu/h^3$: $6\mu V/h^2 - 8\Delta - 8G = \Delta - G$, so $9\Delta = 6\mu V/h^2 - 7G$:

$$p_0 - p_a = \frac{2}{3}\frac{\mu V L}{h^2} - \frac{7}{9}\rho g L \sin\alpha$$

The pressure $p = P - \rho g x \sin\alpha$ is linear in each stretch, because the weight term cancels (as in VF5):

$$0 < x < L: \quad p = p_a + (p_0 - p_a)\frac{x}{L}, \qquad L < x < 2L: \quad p = p_0 - (p_0 - p_a)\frac{x - L}{L}$$

(the second from $P(x) = P(L) + P_{l2} \cdot (-(x - L))$ and $p = P - \rho g x \sin\alpha$).

Volume flux. Using the second stretch with $P_{l2} = \Delta - G$:

$$Q = \frac{Vh}{2} + \frac{h^3}{12\mu}\left(\frac{2}{3}\frac{\mu V}{h^2} - \frac{16}{9}G\right) = \frac{Vh}{2} + \frac{Vh}{18} - \frac{4}{27}\frac{h^3 G}{\mu} = \frac{5}{9}Vh - \frac{4}{27}\frac{\rho g h^3 \sin\alpha}{\mu}$$

Minimum velocity. $Q > 0$ requires $\frac{5}{9}Vh > \frac{4}{27}\frac{\rho g h^3 \sin\alpha}{\mu}$:

$$V > V_{min} = \frac{4}{15}\,\frac{\rho g h^2 \sin\alpha}{\mu}$$

Power Needed to Drive the Belt

The liquid acts on the belt with the shear $\tau = \mu\,\partial v_x/\partial y$ at $y = 0$ (normal $+\bar{e}_y$, from the belt into the liquid):

$$\tau_1 = \mu\left(-\frac{V}{2h} + \frac{P_{l1}}{2\mu}\,2h\right) = -\frac{\mu V}{2h} + P_{l1}h, \qquad \tau_2 = -\frac{\mu V}{h} + \frac{P_{l2}h}{2}$$

$$\tau_1 + \tau_2 = -\frac{3}{2}\frac{\mu V}{h} + h\left(P_{l1} + \frac{P_{l2}}{2}\right) = -\frac{3}{2}\frac{\mu V}{h} - h\left(\frac{\Delta}{2} + \frac{3G}{2}\right)$$

The force of the liquid on the belt (per unit width) is $L(\tau_1 + \tau_2)\bar{e}_x$, negative: the liquid resists. The key's boxed $W = VL(\tau_1 + \tau_2)$ is the power of this force (negative); **the power that the drive must supply is its opposite**:

$$W_{drive} = -VL\,(\tau_1 + \tau_2) = VL\left[\frac{3}{2}\frac{\mu V}{h} + \frac{3}{2}\rho g h \sin\alpha + \frac{h}{2}\,\frac{p_0 - p_a}{L}\right]$$

and with $\Delta = \frac{2}{3}\frac{\mu V}{h^2} - \frac{7}{9}G$ ($\frac{h\Delta}{2} = \frac{1}{3}\frac{\mu V}{h} - \frac{7}{18}Gh$):

$$W_{drive} = VL\left[\frac{11}{6}\,\frac{\mu V}{h} + \frac{10}{9}\,\rho g h \sin\alpha\right]$$

Energy check (computed here, horizontal case $G = 0$): the viscous dissipation of a gap H per unit length is $\int_0^H \mu\,(\partial v_x/\partial y)^2 dy = \mu V^2/H + P_l^2 H^3/(12\mu)$ (the cross term integrates to zero). With $P_{l1} = -\Delta = -\frac{2}{3}\mu V/h^2$ and $P_{l2} = \Delta$: stretch 1 gives $\frac{43}{54}\mu V^2/h$ and stretch 2

gives $\frac{28}{27}\mu V^2/h = \frac{56}{54}\mu V^2/h$; adding them and multiplying by L gives $\frac{99}{54}\mu V^2 L/h = \frac{11}{6}\mu V^2 L/h$, which is W_{drive} for $G = 0$.

Answers and Checks

- Viscous dominance: $(\rho V h/\mu)(h/L) \ll 1$.
- Pressure: $p = p_a + (p_0 - p_a)x/L$ for $0 < x < L$ and $p = p_0 - (p_0 - p_a)(x - L)/L$ for $L < x < 2L$, with $p_0 - p_a = \frac{2}{3}\mu V L/h^2 - \frac{7}{9}\rho g L \sin\alpha$.
- Flux: $Q = \frac{5}{9}Vh - \frac{4}{27}\rho g h^3 \sin\alpha/\mu$; $V_{min} = \frac{4}{15}\rho g h^2 \sin\alpha/\mu$.
- Power supplied: $W = VL\left[\frac{3}{2}\frac{\mu V}{h} + \frac{3}{2}\rho g h \sin\alpha + \frac{h}{2}\frac{p_0 - p_a}{L}\right] = VL\left[\frac{11}{6}\frac{\mu V}{h} + \frac{10}{9}\rho g h \sin\alpha\right]$.
- Dimensions: $\mu V^2 L/h$ per unit width is $ML^{-1}T^{-1} \cdot L^2 T^{-2} \cdot L/L = MLT^{-3}$, a power per unit width; $\rho g h^2/\mu$ is a velocity.
- Limits: for $\alpha = 0$ and $L \to$ any, $V_{min} = 0$ and $Q = \frac{5}{9}Vh$ (the pump works for any $V > 0$); for h fixed and large α the weight of the column stops the pumping when $V < V_{min}$.
- Check against the official key: VF9.pdf gives the same profiles, fluxes, $p_0 - p_a$, Q, V_{min} and pressure distribution (its p is the pressure, in which the weight cancels), and $W = VL(\tau_1 + \tau_2)$, which has the sign of the force on the belt, as explained above.